Declaration

In accordance with University Policy, I certify that:

The attached work submitted for assessment is my own work and that all material drawn from other sources has been fully acknowledged and referenced.

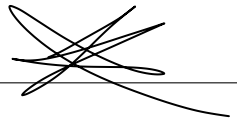
Use of AI tools

Students are permitted to use AI tools for general research purposes and to check and improve the quality of written English in their report. Students are not permitted to use AI tools to generate content.

I have used AI tools in the preparation of this report: **Yes**

If yes, then provide details:

Claude Opus 4.6 (Anthropic) was used throughout the preparation of this proposal as a drafting and editorial tool. Specific uses included: critiquing and restructuring draft sections (particularly the method and aims), verifying that claims attributed to cited sources were faithfully represented in the original works, retrieving and formatting bibliographic entries for sources not already in hand, generating LaTeX code for the Gantt chart and system architecture diagram, and systematic proofreading for grammar, formatting, and IEEE referencing compliance. All research direction, problem formulation, and intellectual content originated from the author in consultation with the supervisor. The author reviewed, edited, and accepted or rejected all AI-generated suggestions.

Signed: 

Date: 13/03/2026

Research Proposal

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Title: Towards Reactive AI-Driven Moving Target Defence: Parameter Analysis and Adversarial Modelling in a Network Simulator

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Degree: Bachelor of Advanced Computer Science (Honours)

1 Introduction

1.1 Moving Target Defence

Moving Target Defence (MTD) is the deliberate creation of asymmetric uncertainty that redirects the cost advantage from the attacker to the defender [1]. The fundamental design principle for developing MTD techniques lies in the decisions for the following three questions: *what to move, how to move, and when to move* [2]. Critically, there is a distinction between proactive MTD (reconfiguration at fixed intervals) and reactive MTD (triggered by specific events such as IDS alerts). This results in a fundamental trade-off. If the interval time is too large, the attacker has sufficient time to complete an attack before the next move. If the interval time is too small, the MTD triggers even when there is no attack on the system, wasting defence resources and degrading performance [3]. This is the core motivation for a reactive MTD that uses IDS feedback to make informed reconfiguration decisions rather than moving blindly.

Zhang (2023) [4] extended the foundational *MTDSim* simulator [5] into *MTDSimTime*, a Discrete-Event Simulation framework that integrates the time domain into the evaluation of combined MTD strategies. Zhang's evaluation demonstrates that network-shuffle-based techniques (IP Shuffle, Complete Topology Shuffle) consistently outperform diversity-based techniques by over 130% on average. However, the work has several limitations that constrain the real-world scalability of its conclusions. First, the attacker model is notably shallow: only one of two originally available attack scenarios was implemented, and the adversary follows a fixed action sequence with no adaptive intelligence, representing the lowest common denominator of attacker sophistication. Second, the MTD operations module was reduced from seven techniques to four, and evaluation was limited to single and pairwise combinations only, leaving open the question of whether further combinations yield further gains.

Tay (2024) [6] extended the *MTDSimTime* framework by producing *MTDShield*, a reinforcement learning module that uses Double Deep Q-Network (DDQN) to automate the selection and timing of MTD operations based on network security metrics. The system processes both static features and temporal metrics through separate neural network

branches before fusing them to produce a set of five Q-values representing possible MTD actions. This work is a meaningful step towards reactive MTD deployment, moving beyond fixed-interval scheduling by basing defensive actions on the observed network state. However, *MTDShield* has notable limitations. Most critically, some metrics used for decision making (such as Host Compromise Ratio or Mean Time to Compromise) are aggregate measures that assume the defender has complete and accurate knowledge of the network state. Further investigation of the input parameters driving MTD decision-making is needed to make the defence truly reactive.

Both works are constrained by aspects of the underlying simulation that remain underexplored. The attacker model lacks adaptive behaviour, the MTD action space remains artificially reduced from its original scope, and the input parameters driving the RL agent's decisions have not been systematically evaluated for their individual contribution to defensive outcomes. These elements require systematic investigation and strengthening as a precursor for further work.

1.2 Intrusion Detection in Moving Target Defence

MTD does not exist in isolation. An Intrusion Detection System (IDS) provides the observation channel through which defenders perceive attacker activity, while MTD provides the action space through which defenders alter the attack surface. The relationship is mutually reinforcing: MTD forces attackers into anomalous behaviours that IDS can more readily detect, while IDS provides the situational awareness needed for intelligent, event-driven reconfiguration rather than costly blind randomisation [7]. Previous works on this simulator [4] characterised IDS and firewalls as static legacy mechanisms that MTD supersedes. This proposal adopts a more nuanced framing, treating IDS as an architectural complement to MTD. A consequence of this integration is that the defender never observes the true state of the network. The defender sees only noisy IDS alerts, subject to missed detections, false alarms, and variable latency [8]. Network defence under these conditions is fundamentally a problem of partial observability, formalised in the literature as a Partially Observable Markov Decision Process (POMDP) [9]. Current work on this simulator does not enforce any such constraint: the RL agent learns from network security metrics directly.

Tay [6] undertook a preliminary investigation into this gap through a sensitivity study that varied the proportion of attacker actions visible to the model during training, simulating the effect of imperfect detection. The study found that model performance degrades below 70% detection visibility, with performance becoming uncorrelated with detection rate at lower sensitivities. However, this approach models IDS as a uniform random masking of attacker actions — a single scalar parameter applied identically across all attack types. Replacing this simplified model with a realistic IDS observation layer is a key stretch goal of this work.

1.3 Optimisation of MTD Operations

Beyond the observation model, prior work has also optimised the reinforcement learning agent's configuration. Tay [6] evaluated DDQN hyperparameters and conducted ablation studies on the *MTDShield* architecture, finding that removing either the static or temporal input module degraded performance by only 6–8%. Dunstall (2025) [10] extended this through multi-objective optimisation using NSGA-II, introducing cost metrics such as downtime ratio and MTD action ratio alongside security metrics. However, both efforts

inherit the full-observability assumptions, constrained action space, and simplified attacker model of the underlying simulator. This project includes systematic parameter exploration but treats it as a component of strengthening the simulation framework rather than an end in itself.

2 Aim

This project aims to strengthen the *MTDShield* simulation framework [4, 6, 10] and systematically evaluate how the AI-MTD module's input parameters, observation configurations, and adversarial conditions affect its defensive performance. Prior work evaluated the RL agent under fixed configurations and a simplified attacker; this project addresses those constraints. Specifically, this project will:

- **Develop a more sophisticated adversarial module** that exercises the AI-MTD agent beyond the capabilities of the existing baseline attacker, enabling non-trivial evaluation of defensive strategies.
- **Systematically explore AI-MTD input parameters and configurations** to identify which parameter regimes most meaningfully affect defensive outcomes.
- **Analyse and characterise the relationship between configuration choices and performance**, offering empirical insight into not only which settings improve outcomes but why.
- **(Stretch) Design and integrate an IDS observation module** that replaces the agent's current full-observability assumption with a realistic, partial-observation interface, and evaluate agent performance under degraded observation conditions.

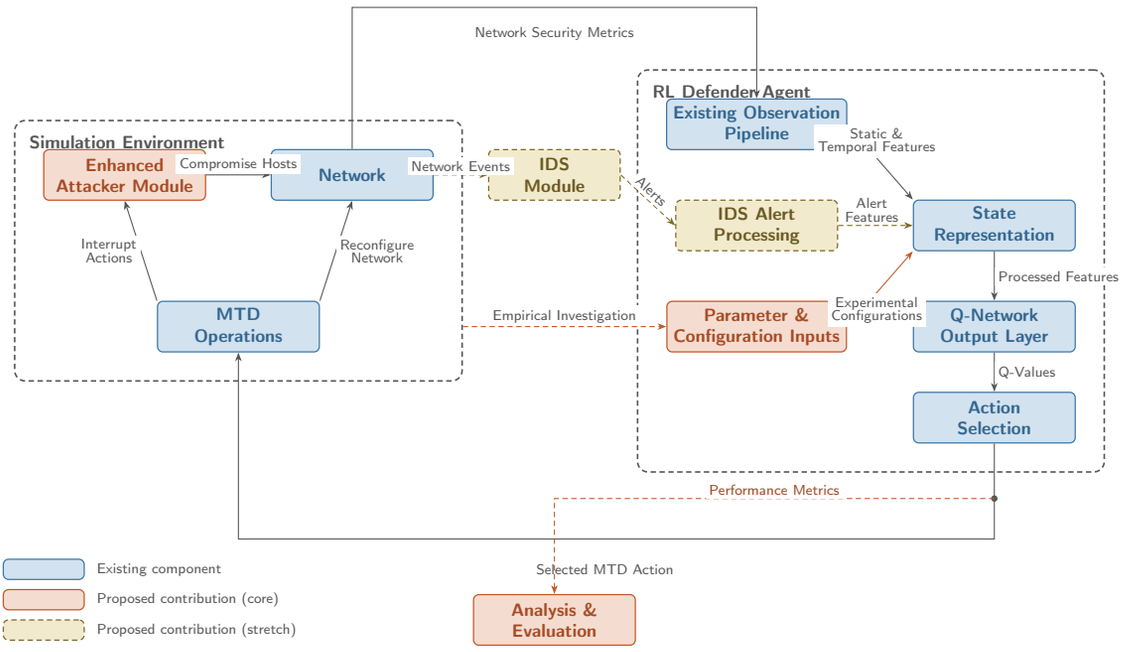


Figure 1: Overview of proposed contributions to the simulation architecture.

3 Method

This section outlines the approach for achieving each aim identified above, broken into sequential and overlapping phases. A consolidated timeline is presented in Section 3.5.

3.1 Codebase Comprehension and Baseline Establishment

A prerequisite to any experimental contribution is a thorough understanding of the existing *MTDShield* codebase and its antecedents [4, 6]. This phase, conducted in parallel with the literature review (March–April 2026), involves auditing the simulator architecture, reproducing key results from prior work to establish baseline performance figures, and documenting the codebase to support future extension. Baseline reproduction serves as the completion criterion: the phase is complete when prior results have been independently verified within an acceptable margin.

3.2 Adversarial Model Enhancement

The existing attacker follows a fixed action sequence with no adaptive behaviour, and only one of two originally available attack objectives was implemented [4]. This phase strengthens the adversarial module by first reintegrating the second attack objective, then introducing more sophisticated attack behaviours informed by mapping the simulator's action space to the MITRE ATT&CK framework to identify coverage gaps. New attack scenarios will implement multi-stage kill chains combining reconnaissance, lateral movement, and privilege escalation, calibrated so that the current AI-MTD policy exhibits measurable degradation relative to the baselines from Section 3.1. This phase is estimated at approximately two months (May–June 2026).

3.3 Parameter Exploration and Evaluation

With an enhanced attacker in place, this phase systematically varies the AI-MTD module's inputs and configurations across three categories: observation features, action-space configuration, and training hyperparameters (reward shaping, discount factor, exploration schedule). For each category, individual parameters will be varied against a held baseline (ablation studies), followed by interaction effects between promising combinations. Where parameter changes produce significant performance differences, further analysis will seek to explain *why* — for example, examining which observation features correlate most strongly with effective action selection. This phase runs concurrently with the later stages of adversarial enhancement (late May–late July 2026).

3.4 IDS Integration (Stretch Goal)

This phase is contingent on satisfactory progress in the preceding phases and is scoped as a proof of concept. The objective is to replace the RL agent's current full-observability assumption with an IDS module that intercepts simulated attacker events and produces alerts with configurable detection probability, false positive rate, and detection latency. The agent's observation space would be rewired to receive these alerts rather than ground-truth network state, and retrained under partial-observability conditions. The parameter exploration from Section 3.3 would then be extended to evaluate performance as IDS

fidelity is varied, building on Tay's preliminary sensitivity analysis [6]. If reached, this phase would occupy approximately August–September 2026.

3.5 Timeline

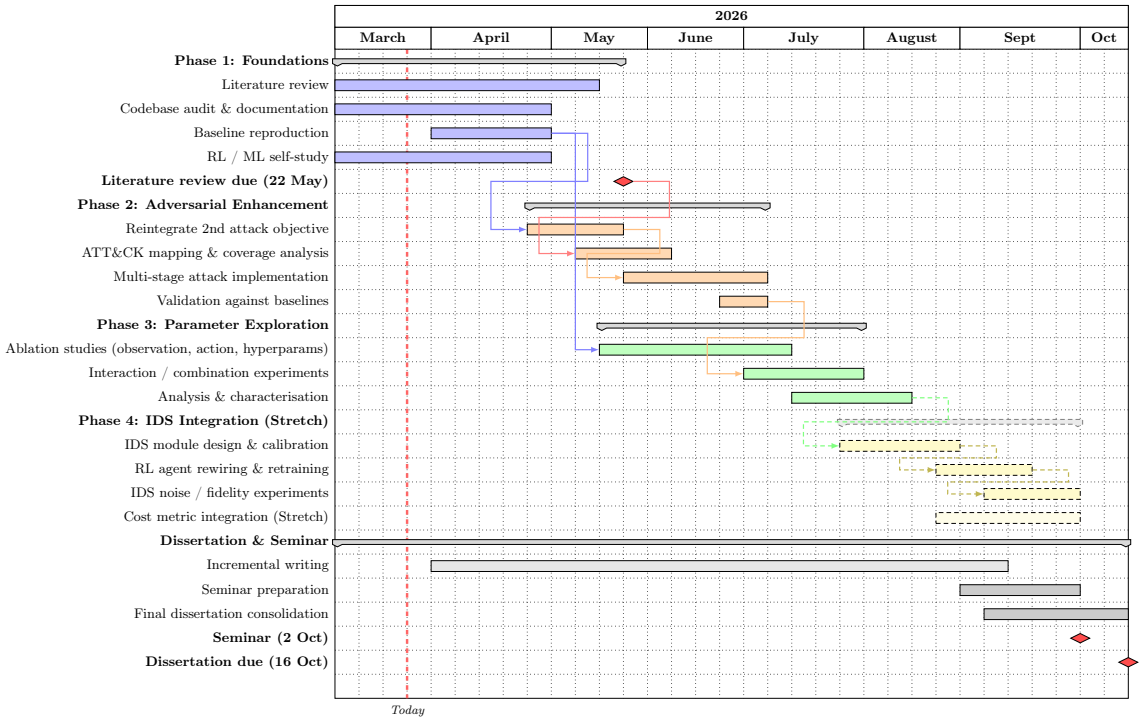


Figure 2: Project timeline. Dashed elements indicate stretch goals.

4 Project Requirements

The project will be conducted in Python 3, expanding on the *MTDShield* [6] framework and its dependencies. All prior codebases are publicly available on GitHub. Training of reinforcement learning models will be performed on commodity hardware; no specialised GPU infrastructure is anticipated. The simulator generates synthetic network data and does not involve human subjects, personal data, or external datasets; no ethics approval is required. Writing and code production will be assisted through the use of Claude Opus 4.6. No IP or access restrictions have been identified.

References

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O.K.

Good overview. Consider the research questions you're addressing and also the processes you will use to validate the simulations.

Well done.